
Probability of Matching for Batch Multi-Objective Bayesian Optimization

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Abstract

In batch multi-objective Bayesian optimization (MOBO), it is often desirable to identify the whole Pareto optimal set, especially when considering the complicated interplay between different design criteria and constraints. This poses unique challenges in acquiring batches of both high quality and diversity to cover the Pareto front. We propose a novel acquisition strategy, Probability of Matching (POM), which evaluates both batch candidate quality and diversity by explicitly capturing the likelihood that all batch points are Pareto optimal, and the probability that they collectively cover the full Pareto set. To estimate the coverage probability and promote diversity, we incorporate new sampling principles conditioned on Pareto possibility, resulting in our new POM-guided batch MOBO method. Across synthetic benchmarks and real-world tasks, our method consistently outperforms state-of-the-art baselines on standard MOBO metrics as well as a new design-space coverage metric, Expected Minimum Distance (EMD), with comparable computational efficiency.

1. Introduction

Bayesian optimization (BO) is a powerful framework for optimizing unknown or uncertain objective functions that are complex (non-linear, non-convex, non-stationary) and expensive to evaluate in terms of both time and cost (Tu et al., 2022; Garnett, 2023; Ahmadianshalchi et al., 2024; Wang et al., 2024). BO operates by sequentially query-

ing objective function values for selective input, due to the “black-box” nature of the problem setting. To achieve the desired sampling and computation efficiency, it models the objective function probabilistically, constructs an acquisition function considering model uncertainty, and uses this acquisition function to iteratively select new points in the design space to query. The probabilistic model is updated with new observations at each iteration.

When multiple objectives are involved, BO becomes more complex compared to those targeting single-objective problems. Instead of identifying a single global optimum, the goal here is to approximate the non-dominated Pareto optimal set. In this context of multi-objective BO (MOBO), acquisition functions must evaluate the quality of samples relative to the unknown Pareto front. A widely used approach is by the Expected Hypervolume Improvement (EHVI; Emmerich et al. (2006)), which measures improvement in the hypervolume dominated with respect to a reference point. However, the effectiveness of EHVI is sensitive to the choice of the reference point, which can bias sampling toward specific regions of the objective space (Auger et al., 2009). Park et al. (2023) developed BOTied adopting a cumulative distribution function (CDF) based metric that estimates the number of objective values each candidate dominates. Unlike hypervolume based criteria, this CDF metric does not require a reference point and is computationally more efficient in high-dimensional objective spaces. However, it can bias searching toward locally promising regions rather than the true Pareto front globally once the surrogate is not equally represented across the design space. As a consequence, the method has shown myopic behavior with oversampling evaluations that concentrate excessively in a local Pareto optimal regions.

Empirically, batch BO can partially address myopic behavior as it mitigates diversity limitations by suggesting multiple candidates simultaneously, reducing the risk of oversampling in a single localized region (Wang & de Freitas, 2014; Auger et al., 2009). For MOBO, batch selection is inherently more challenging due to several factors: the acquisition function must account for multiple objectives; it must capture interactions among batch points to promote diversity and reduce redundancy; and it must carefully balance exploration of the design space with exploitation of known high-performing regions.

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In this paper, we propose a novel acquisition function by introducing a probabilistic framework with the Probability of Matching (POM), which can guide batch MOBO to better balance the trade-off between exploiting regions with known high performance and ensuring diversity among batch points. This new acquisition function captures the likelihood that an acquired batch matches the true Pareto optimal solutions. To estimate this POM effectively, we factorize it into two components: (i) *Pareto Optimality*, the probability that all batch points belong to the Pareto set, and (ii) *Coverage*, the probability that the batch collectively covers all Pareto optimal solutions. By jointly estimating these two components during batch MOBO, our approach dynamically balances query candidate quality with solution diversity, enabling more effective multi-objective batch acquisition.

Unlike prior work (Emmerich et al., 2006; Park et al., 2023), in which *Pareto Optimality* and *Coverage* are typically characterized only through an informal, positively correlated notion of proximity to the true Pareto front, we provide precise probabilistic definitions that links *Pareto Optimality* and *Coverage*, to the probability of acquiring the global Pareto front.

Specifically, we define *Pareto Optimality* as the probability that a candidate is non-dominated by other candidates, which is conceptually analogous to the Probability of Optimality introduced in single-objective BO (Hennig & Schuler, 2012). To estimate *Coverage*, we employ both hypervolume and CDF indicators. The rationale for interpreting these indicators as coverage estimators is discussed in detail in Section 3.1. To further mitigate the commonly observed myopic behavior, we adopt a conditional sampling strategy that ensures samples are drawn from the Pareto front possible regions, rather than being generated solely from the current surrogate model as in standard estimation methods.

Building on these probabilistic estimates, we develop POM-based batch MOBO, for which we derive two new acquisition strategies: POM with Hypervolume (POMH) and POM with CDF (POMC). These methods enable faster approximation of Pareto optimal solutions (improving *Pareto Optimality*) while simultaneously yielding higher quality Pareto fronts (enhancing *Coverage*), without excessive evaluation budget being wasted on local oversampling with conditional sampling mechanism.

To validate the effectiveness of our POM framework, we evaluate our POM methods across a range of challenging multi-objective optimization tasks. Compared to baseline methods, including EHVI and the diversity-aware Quantile Stein Variational Gradient Descent (QSVGD; Gong et al. (2019)), our POM batch MOBO consistently achieves superior rediscovery performance across standard evaluation metrics, including hypervolume, inverted generational distance (IGD; Ishibuchi et al. (2015)) and other widely used

benchmarks for MOBO. We also introduce a new metric, Expected Minimum Distance (EMD), to evaluate coverage in the design space, where our method again outperforms existing baselines. In addition, we conduct a materials discovery case study, where POM demonstrates superior performance with consistently better rediscovery ratio.

2. Multi-Objective Bayesian Optimization

Background. Multi-objective Bayesian optimization (MOBO) can be formulated by considering optimizing an array of multiple objective functions with respect to the decision or design variables in a well-defined design space, $\mathcal{X} \in \mathcal{X}$:

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x})), \quad (1)$$

where $f_1(\cdot), \dots, f_m(\cdot)$ are m design objectives defined similarly as in single-objective Bayesian optimization (SOBO). Unlike SOBO, which aims to identify a single global or local optimum, the objective of MOBO is to generate a set of designs that capture the best trade-offs across competing objectives, forming the *Pareto front*. Denote the image of the m objective functions as $\mathcal{Y} \subset \mathbb{R}^m$, the Pareto front is defined as $\mathcal{Y}^* = \{\mathbf{y} \in \mathcal{Y} : \nexists \mathbf{y}' \text{ s.t. } \mathbf{y}' \succ \mathbf{y}\}$ if the goal is to maximize each objective. Here, $\mathbf{y} \succ \mathbf{y}'$ reads as $\mathbf{y}$ dominates $\mathbf{y}'$ in the context of maximization for all m design objectives, meaning that $\forall i \leq m, y_i \geq y'_i$ and $\exists j \leq m, y_j > y'_j$. In the design space, the pre-image of $\mathcal{Y}^*$ is denoted by $\mathcal{X}^*$, called the *Pareto optimal set*.

Batch MOBO extends the standard sequential MOBO by selecting a set of $q > 1$ query design points per iteration, enabling parallel evaluation of these multiple expensive black-box objectives. The goal is still to approximate the Pareto optimal set $\mathcal{X}^*$, but evaluations are performed in batches of size q :

$$\mathbf{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}\} \subseteq \mathcal{X}. \quad (2)$$

At each iteration n , typical implementations of batch MOBO select the batch $\mathbf{X}$ by jointly maximizing a batch acquisition function $\alpha(\mathbf{X} \mid \mathcal{D}_n)$, where $\mathcal{D}_n = \{(\mathbf{x}_i, \mathbf{f}(\mathbf{x}_i))\}_{i=1}^n$ is the set of observed data. The batch selection is by:

$$\mathbf{X} = \arg \max_{\mathbf{X}' \subseteq \mathcal{X}, |\mathbf{X}'|=q} \alpha(\mathbf{X}' \mid \mathcal{D}_n). \quad (3)$$

One widely adopted batch acquisition function in MOBO is qEHVI, which generalizes EHVI to the batch setting by jointly approximating the hypervolume improvement of the entire batch (Emmerich et al., 2006). Assuming $A \subset \mathcal{Y}$ is a finite set of objective vectors, the hypervolume indicator $\mathcal{H}(A)$ is defined as the Lebesgue measure of the region in the objective space dominated by at least one element of A

and bounded above by a reference point $\mathbf{r} \in \mathbb{R}^m$, which is dominated by all elements in the Pareto front:

$$\mathcal{H}(A) = \text{Vol}(\{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y} \succ \mathbf{r}, \exists a \in A : a \succ \mathbf{y}\}) \quad (4)$$

where $\text{Vol}(\cdot)$ quantifies the size of a set of the m -dimensional objective space.

Given a batch of q input points $\mathbf{X}$, the qEHVI acquisition function evaluates the expected increase in the dominated hypervolume over the current Pareto front set $\mathcal{Y}_n^*$:

$$\alpha_n^{\text{qEHVI}}(\mathbf{X} \mid \mathcal{D}_n) := \mathbb{E}_{\mathbf{Y} \mid \mathcal{D}_n, \mathbf{X}} [\mathcal{H}(\mathcal{Y}_n^* \cup \mathbf{Y}) - \mathcal{H}(\mathcal{Y}_n^*)], \quad (5)$$

where $\mathbf{Y} = \mathbf{y}^{(1:q)}$ is the (random) vector of objective values for points $\mathbf{x}^{(1:q)}$ under the Gaussian process (GP; [Rasmussen \(2003\)](#)) surrogate model, and the expectation is taken over the joint posterior distribution of $\mathbf{f}(\mathbf{X})$ conditioned on current observations $\mathcal{D}_n$.

In practice, this expectation is often approximated using Monte Carlo integration by sampling $\mathbf{Y}$ from the posterior and computing the average hypervolume improvement. In MOBO, qEHVI can be biased towards certain regions, depending on the shape of the Pareto front and the position of the reference point ([Auger et al., 2009](#)). For example, if the reference point is far from the Pareto front, qEHVI may prioritize extreme solutions on the Pareto front ([Tian et al., 2016](#)). This bias occurs because extreme solutions are likely to yield the greatest hypervolume improvement. However, the ultimate goal of MOBO is to collect evaluation samples across the entire Pareto front to support more informed design decisions. This calls for new decision-making strategies that ensure better coverage performance during acquisition.

Related Work Several existing methods have been developed to improve MOBO performance through better coverage of the Pareto front, but most of these operate directly in the objective space. For example, Expected MaxiMin Improvement (EMMI; [Olofsson et al. \(2018\)](#)) enhances Pareto front coverage by prioritizing improvement in unexplored regions of the objective space. Similarly, Inverted Generational Distance with Noncontributing Solutions (IGDNS; [Tian et al. \(2016\)](#)) aims to improve coverage by directly minimizing the IGD metric ([Zhou et al., 2006](#)). These methods are effective when Pareto optimal solutions are densely clustered in a specific region in the design space. However, these approaches struggle to capture the full Pareto front when the solutions are widely dispersed across the design space. Implementing diversity improvement directly in the objective space also presents several challenges: 1) Invalidity: It is unclear whether any candidates actually exist in the regions of the objective space being targeted; 2) Concept mixture: Pareto front coverage is jointly determined by design space diversity and optimal region exploitation.

Design space diversity establishes the global backbone of the Pareto front, while optimal region exploitation governs how densely and uniformly solutions populate this structure. Treating Pareto front coverage purely as an objective space diversity problem is therefore an oversimplification, as it may encourage the search to concentrate on redistributing solutions within the current backbone rather than expanding the global Pareto front structure; 3) Misalignment with optimization goals: Diversity improvement strategies in the objective space may propose queries that are inconsistent with the primary optimization objective, such as directions that may decrease the hypervolume metric, thereby potentially degrading overall optimization performance.

Given the limitations of promoting diversity in the objective space, enhancing the diversity of Pareto optimal solutions in the design space presents a more reliable alternative. This is because: 1) the feasible region is explicitly defined, eliminating concerns about solution validity; 2) the diversity modeling focuses only on constructing the backbone of the Pareto front, while the population of solutions within this backbone is considered as another independent task that be governed by optimal region exploitation; 3) promoting diversity in the design space does not compromise solution quality, as there is no inherent preferential direction within the feasible domain; 4) and moreover, diversity estimation in the design space does not introduce additional sensitivity to observational noise if the diversity estimation is computed purely from the distribution of the candidates, thereby preserving the robustness of the approach.

However, not many related works have taken into account the diversity of Pareto optimal solutions when developing MOBO. Here, before we present our new POM-based batch MOBO, we first consider a few modified strategies. The first one QSVGD ([Gong et al., 2019](#)), which has been developed to increase the exploration ability in SOBO. We extend the original implementation into batch MOBO and still refer to it as QSVGD throughout the paper. QSVGD incorporates batch entropy to enhance the diversity of queried samples. This formulation, though effective for global exploration, does not directly target diversity of the solutions. Therefore, we use this multi-objective QSVGD approach solely as a baseline for comparison with our solutions' diversity improvement strategy introduced later. The acquisition function for QSVGD is defined as: $\alpha_n^{\text{QSVGD}}(\mathbf{X} \mid \mathcal{D}_n) :=$

$$\mathbb{E}_{\mathbf{Y} \mid \mathcal{D}_n, \mathbf{X}} [\mathcal{H}(\mathcal{Y}_n^* \cup \mathbf{Y}) - \mathcal{H}(\mathcal{Y}_n^*)] + \eta \mathcal{H}(\mathbf{X}), \quad (6)$$

where $\mathcal{H}$ is the entropy of the distribution over the batch samples and η is the hyperparameter to balance the batch quality and diversity.

Besides QSVGD, other methods such as Pareto Optimal Thompson Sampling (qPOTS; [Renganathan & Carlson \(2023\)](#)) and Diversity Guided Evolutionary Multi-Objective

Optimization (DGEMO; Neumann et al. (2019)) also incorporate design space diversity. In these approaches, diversity is primarily quantified through distances between solutions in the design space. However, unlike entropy based measures, distance is not an ideal diversity metric because distance itself is not uniquely defined, making it difficult to determine which formulation is most suitable for characterizing diversity.

A more fundamental limitation of all these diversity considered methods is that they implicitly assume diversity should always be encouraged during optimization. In practical black-box problems, however, the true structure of the objective function is unknown a priori, making it impossible to determine the appropriate level of diversity before sufficient exploration has occurred. For instance, when the underlying objective is effectively unimodal, aggressively promoting diversity, as in methods such as QSVGD, can significantly reduce optimization efficiency and solution quality. Therefore, an effective optimization strategy should adaptively match the unknown objective structure, rather than imposing diversity or exploitation preferences in advance.

3. Batch MOBO with Probability of Matching

3.1. Probabilistic Pareto Set Matching

We propose a novel batch acquisition strategy for MOBO that optimizes a new metric, POM, to guide batch selection. This metric measures the probability that the selected batch is identical to the true Pareto optimal set by quantifying the likelihood that an acquired batch not only contains high-quality solutions, but also closely approximates the entire Pareto optimal set.

We model and optimize the probability that the acquired batch $\mathbf{X}$ approximates the true Pareto optimal set $\mathcal{X}^*$ both in *Pareto Optimality* and *Coverage*. *Pareto Optimality* is defined as $\forall \mathbf{x}' \in \mathbf{X}, \forall \mathbf{x} \in \mathcal{X}, \text{s.t. } \mathbf{f}(\mathbf{x}') \not\preceq \mathbf{f}(\mathbf{x})$, which is equivalent to $\mathbf{X} \subseteq \mathcal{X}^*$. *Coverage* is defined as $\forall \mathbf{x} \in \mathcal{X}, \exists \mathbf{x}' \in \mathbf{X}, \text{s.t. } \mathbf{f}(\mathbf{x}) \preceq \mathbf{f}(\mathbf{x}')$, which is equivalent to $\mathcal{X}^* \subseteq \mathbf{X}$.

Thus, the probability that the acquired batch $\mathbf{X}$ approaches the true Pareto optimal set is

$$\begin{aligned} P(\mathbf{X} = \mathcal{X}^*) &= P(\mathcal{X}^* \subseteq \mathbf{X}, \mathbf{X} \subseteq \mathcal{X}^*) \\ &= P(\mathbf{X} \subseteq \mathcal{X}^*)P(\mathcal{X}^* \subseteq \mathbf{X} \mid \mathbf{X} \subseteq \mathcal{X}^*), \end{aligned} \quad (7)$$

which quantifies the likelihood that the selected batch $\mathbf{X}$ matches exactly the true Pareto optimal set.

As in (7), we factorize this “matching” event $\mathbf{X} = \mathcal{X}^*$ by estimating the joint probability of $\mathcal{X}^* \subseteq \mathbf{X}$ and $\mathbf{X} \subseteq \mathcal{X}^*$: $\mathbf{X} \subseteq \mathcal{X}^*$ is the event of $\mathbf{X}$ belonging to the Pareto optimal set and $\mathcal{X}^* \subseteq \mathbf{X}$ is the event that all optimal solutions are contained within $\mathbf{X}$.

In continuous settings with infinitely optimal solutions, the exact matching event $\mathbf{X} = \mathcal{X}^*$ is almost impossible to occur when pursuing arbitrarily high accuracy, since the batch size q is always finite. To bridge the theoretical gap between infinitely optimal solutions and a limited batch size, we treat the matching probability as a density distribution defined over the space of finite q -sequences on $\mathcal{X}$. For example we draw an analogy between the POM in MOBO and the probability of optimality $P(\mathbf{x} = \mathbf{x}^*)$ in single objective BO. Given an objective function like $f(\mathbf{x}) = (\mathbf{x} - \pi)^2$, we can never sample the exact value π in reality. However, this does not mean $P(\mathbf{x} = \mathbf{x}^*)$ is permanently zero or an invalid estimator; in fact, many acquisition functions, such as entropy-based methods, are developed based on this optimizer distribution $P(\mathbf{x} = \mathbf{x}^*)$. To address this 0 probability issue in real case, we usually convert $P(\mathbf{x} = \mathbf{x}^*)$ to $P(\|\mathbf{x} - \mathbf{x}^*\| < \epsilon)$ as $\epsilon \rightarrow 0$. This builds a bridge between theoretical matching and real-world convergence using a tolerance level ϵ . We apply a similar explanation to our POM. In practice, we consider $P(\|\mathbf{X} - \mathcal{X}^*\| < \epsilon)$ with $\epsilon \rightarrow 0$ as a theoretical substitute. For Pareto optimality, we evaluate whether a candidate batch falls within a hypersphere centered at the true Pareto optimal solution with radius ϵ . The coverage can then be defined as the percentage of the design space dominated by this hypersphere. This substitution ensures that, given an accuracy expectation ϵ , the POM remains non-zero for a continuous Pareto front, eliminating the need for an infinite q to match the true Pareto front.

Pareto optimality In the batch setting, assuming independence across candidate points, the overall Pareto optimality probability is defined as the product of the individual probabilities,

$$\begin{aligned} P(\mathbf{X} \subseteq \mathcal{X}^*) &= P(\forall \mathbf{x}' \in \mathbf{X}, \forall \mathbf{x} \in \mathcal{X}, \mathbf{f}(\mathbf{x}') \not\preceq \mathbf{f}(\mathbf{x})) \\ &= \prod_{\mathbf{x}' \in \mathbf{X}} P(\mathbf{f}(\mathbf{x}') \not\preceq \mathbf{f}(\mathbf{x})). \end{aligned} \quad (8)$$

Indicator (8) encourages the selection of non-dominated solutions. Candidates with extreme objective values naturally attain high scores under the *Pareto optimality* indicator, as their probability of being dominated by other solutions is negligible. For instance, a candidate achieving the optimum value in a single objective cannot be dominated and therefore receives the maximum Pareto optimality score. Overall, the Pareto optimality indicator guides the construction of the boundary of the Pareto front.

Coverage Selecting the coverage indicator is inherently challenging, as verifying the event $\forall \mathbf{x} \in \mathcal{X}, \exists \mathbf{x}' \in \mathbf{X}, \text{s.t. } \mathbf{f}(\mathbf{x}) \preceq \mathbf{f}(\mathbf{x}')$ would require evaluating acquisitions over the entire candidate space, which is not possible in practice. Thus, for each batch $\mathbf{X}$, instead of verifying whether it dominates all candidates, we use the percentage of all candidates dominated by the batch as an estimator of

coverage probability:

$$P(\mathcal{X}^* \subseteq \mathbf{X}) = P_{\mathbf{x} \in \mathcal{X}} \left(\exists \mathbf{x}' \in \mathbf{X} \text{ s.t. } \mathbf{f}(\mathbf{x}) \preceq \mathbf{f}(\mathbf{x}') \right) \\ \approx \frac{1}{\mu(\mathcal{X})} \int_{\mathcal{X}} \mathbf{1} \left(\exists \mathbf{x}' \in \mathbf{X} \text{ s.t. } \mathbf{f}(\mathbf{x}) \preceq \mathbf{f}(\mathbf{x}') \right) d\mu(\mathbf{x}), \quad (9)$$

where μ is a measure defined on the design space $\mathcal{X}$. If a counting measure is adopted for μ and (9) is estimated via Monte Carlo sampling, the resulting estimator can be interpreted as an empirical CDF, which has been directly used as a MOBO acquisition indicator in previous work (Park et al., 2023). Another widely used coverage estimator is the hypervolume defined in (4). Comparing (4) and (9), we observe that hypervolume is also defined based on the coverage concept but in the objective space with respect to a reference point. Beyond this definition similarity, we further emphasize the respective advantages and limitations of these two coverage indicators in Section A.1.

Regarding the coverage of the Pareto optimal set, batch MOBO is generally more effective than sequential counterparts, as a larger batch size naturally leads to higher $P(\mathcal{X}^* \subseteq \mathbf{X})$. On the other hand, excessively large batches may reduce $P(\mathbf{X} \subseteq \mathcal{X}^*)$. Also, such large batch sizes incur substantial costs due to the evaluation of numerous non-Pareto optimal solutions, resulting in inefficient utilization of the evaluation budget. Therefore, simply increasing the batch size should not be viewed as a universal solution for improving coverage. Another drawback of using a large batch size is the increased computational cost (Binois et al., 2025). As the batch size q grows, the number of possible batch combinations increases superlinearly, leading to significantly higher complexity in acquisition optimization and evaluation. Instead, we need a principled methodology that balances $P(\mathcal{X}^* \subseteq \mathbf{X})$ and $P(\mathbf{X} \subseteq \mathcal{X}^*)$ to achieve effective and diverse sample queries.

3.2. Pareto optimal region sampling

To guide the next batch selection after n acquired evaluations, we adopt the following formulation to estimate POM,

$$P(\mathbf{X}_{f^*}^* = \mathbf{X}_{n+q}) := \quad (10) \\ P(\mathbf{X}_{f^*}^* = \mathbf{X}_{n+q} | \mathbf{X}_{f^*}^* = \mathbf{X}_n) P(\mathbf{X}_{f^*}^* = \mathbf{X}_n) \\ + P(\mathbf{X}_{f^*}^* = \mathbf{X}_{n+q} | \mathbf{X}_n \neq \mathbf{X}_{f^*}^*) P(\mathbf{X}_n \neq \mathbf{X}_{f^*}^*).$$

Since the optimization objective is to iteratively advance the current Pareto front toward the ground-truth Pareto front, we assume that the current front remains strictly dominated by the ground truth, i.e., there exists non-zero improvement potential at every iteration. Under this assumption, the first term can be considered as constant and can therefore be ignored during acquisition function estimation, as the condition $\mathbf{X}_{f^*}^* = \mathbf{X}_n$ would imply that no further improvement

is possible. For the second term we decompose the condition $\mathbf{X}_n \neq \mathbf{X}_{f^*}^*$ into two cases, $\mathbf{X}_{f^*}^* \setminus \mathbf{X}_n \neq \emptyset$ and $\mathbf{X}_{f^*}^* \setminus \mathbf{X}_n = \emptyset$. In the first case, the future matching probability $P(\mathbf{X}_{f^*}^* = \mathbf{X}_{n+q})$ is 0. Therefore, only the second case contributes to the estimation of $P(\mathbf{X}_{f^*}^* = \mathbf{X}_{n+q})$, and optimizing POM reduces to maximizing:

$$P(\mathbf{X}_{f^*}^* \setminus \mathbf{X}_n = \mathbf{X}_q | \mathbf{X}_n \subset \mathbf{X}_{f^*}^*) P(\mathbf{X}_n \subset \mathbf{X}_{f^*}^*). \quad (11)$$

Figure 1 shows how the estimation is done condition on $\mathbf{X}_n \subset \mathbf{X}_{f^*}^*$. Each box, or combination of boxes, represents a distinct possible outcome of event $\mathbf{X}_n \subset \mathbf{X}_{f^*}^*$. For instance, sampling from the top-right region indicates that none of the points in $\mathbf{X}_n$ lies on the Pareto front; the probability of this outcome is 0.03, which can be computed directly as box CDF from the posterior mean μ , variance σ^2 , and the corresponding box bounds l and u . The expected location associated with each box can also be computed by $\mu + \sigma \frac{\phi(\alpha) - \phi(\beta)}{\Phi(\beta) - \Phi(\alpha)}$, where $\alpha = \frac{u - \mu}{\sigma}$, $\beta = \frac{l - \mu}{\sigma}$. We aggregate all expected locations and associate them with their corresponding selection probabilities to construct a multinomial distribution. Samples drawn from this distribution are then used to estimate the indicator values.

This box-decomposition perspective also offers an alternative explanation of the batch-size effect. Sequential optimization may fail to achieve optimal performance because a single candidate cannot represent all possible outcomes of the event $\mathbf{X}_n \subset \mathbf{X}_{f^*}^*$. For example, in Figure 1, a batch of at least size two is required to simultaneously dominate Pareto front 1 and 4 in order for the event $\mathbf{X}_{2,3} \subset \mathbf{X}_{f^*}^*$ to occur. Under sequential sampling, the condition $\mathbf{X}_{2,3} \subset \mathbf{X}_{f^*}^*$ is never considered, leading to a systematic bias toward representable conditions (regions). Conversely, when the batch size becomes very large, multiple candidates may fall into the same region, and such overlapping evaluations provide limited additional information, resulting in reduced sampling efficiency.

3.3. Complexity analysis

The computational complexity cannot be characterized by a single closed-form expression, as it is primarily governed by the number of regions defined in Section 3.2, which in turn depends on the number of currently Pareto optimal solutions. Following a classical strategy (Bentley et al., 1978), we estimate the number of regions as a function of the total number of acquisitions n and the number of objectives m , yielding an upper bound of $O(n(\ln n)^{m-1})$. $n(\ln n)^{m-1}$ represents the worst case when all of the previous acquisitions are Pareto optimal. The remaining computational cost during the sampling process is dominated by the evaluation of truncated multivariate Gaussian CDFs. Leveraging recent research (Cao & Katzfuss, 2025), this cost is reduced to $O(m)$. Once the regions and their expected lo-

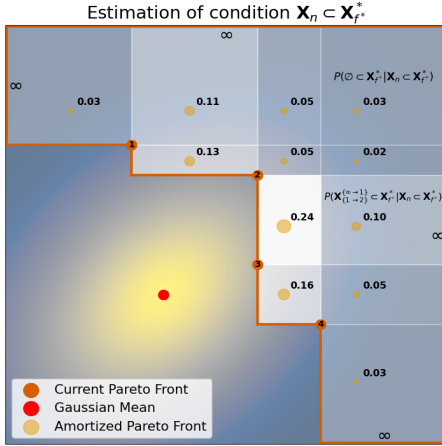


Figure 1. Amortized Pareto front estimation for single candidates in a two-dimensional objective space. The orange frame denotes the valid sampling region; beige regions indicate amortized Pareto front locations, with radii proportional to their sampling probabilities. The white frame shows the region boundaries. Black numbers report the CDF values, representing the probability mass assigned to each region.

cations are determined, all subsequent computations have closed-form expressions and can be computed analytically. Specifically, computing the hypervolume indicator incurs a cost of $O((2^q - 1)m)$, while both the CDF indicator and Pareto optimality have scales as $O(qm)$. Since sampling and indicator computation are performed sequentially, the overall per-iteration complexity is $O(\max\{qm, (2^q - 1)m\})$ in the early stages, and becomes $O(mn(\ln n)^{m-1})$ in later iterations.

4. Empirical results

4.1. Acquisition comparison

We first illustrate how the choice of indicator affects POM’s ability to capture a complex disconnected Pareto front in both the design and objective spaces with a 2D Gaussian mixture (GM) problem, where two modes form partial Pareto fronts and the third mode corresponds to a local Pareto optimal region. This problem enables us to assess whether POM can recover the complete Pareto front despite discontinuities, the presence of potentially trapping local Pareto optimal regions, and the overall complexity of the landscape.

As Figure 2 shows, both POM methods recover the whole Pareto front. However they show different acquired candidates have different behavior. As illustrated in Figure 2d, the hypervolume indicator yields a more uniformly distributed Pareto front, including in extreme value regions that are difficult to reach in the design space. This behavior arises because the hypervolume indicator assigns uniform impor-

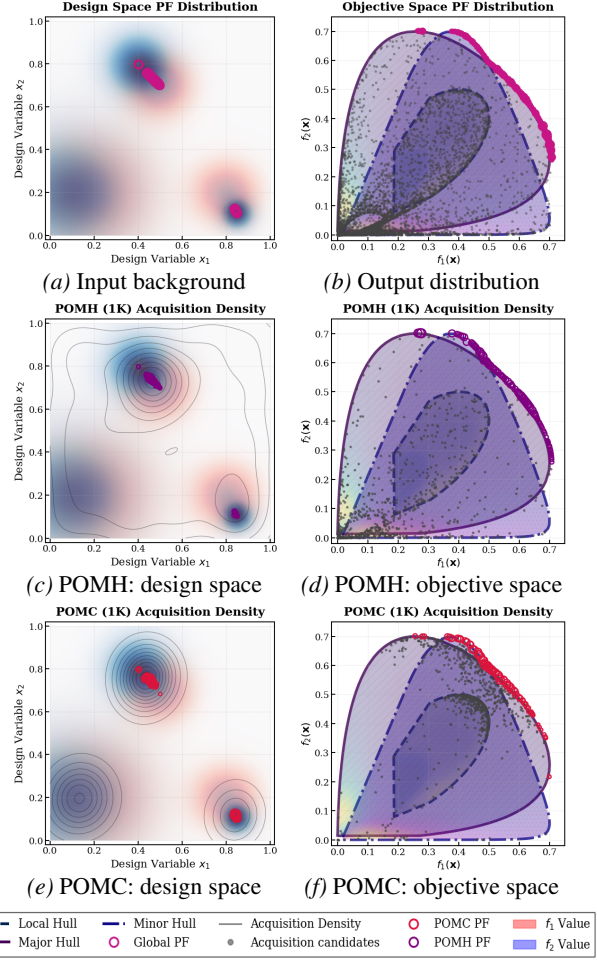


Figure 2. Acquisition distribution comparison between Hypervolume and CDF indicators: (a) Design space ground truth landscape; (b) True objective value distribution for 1M random samples; Hypervolume based POM (POMH) acquisition distribution in design space (c) and objective space (d); CDF based POM (POMC) acquisition distribution in design space (e) and objective space (f).

tance to objective values regardless of the difficulty of attaining them, as discussed in Section 3.1. In contrast, the CDF based indicator exhibits a clear preference toward regions of the Pareto front that are denser and supported by richer data as Figure 2f shows. In data sparse regions, such as the right side in Figure 2f, its Pareto coverage is noticeably weaker than that achieved by the hypervolume indicator.

From the design space perspective, the hypervolume indicator encourages broader exploration, as reflected by the more dispersed acquisition density contours in Figure 2c. Conversely, Figure 2e indicates that the CDF indicator concentrates acquisition mass in regions near the optimal regions since candidate points in these regions are more likely to dominate a larger subset of objective values.

Overall, effectiveness of hypervolume and CDF depends on the intended optimization objective. In most of the applica-

tions, a uniformly distributed Pareto front is preferred, in which case the hypervolume indicator is a suitable choice. Meanwhile when the objective is to recover a Pareto front that reflects its underlying density, or when operating in a discrete design space, the CDF may be more appropriate.

4.2. Baseline benchmark

We have performed experiments in different setups to investigate performance trends of batch MOBO strategies with different batch sizes. To emphasize the importance of reliable coverage estimation of $P(\mathcal{X}^* \subseteq \mathcal{X})$, we focus on the MOBO problems that have multiple Pareto optimal regions in both design space and objective space. For problems with only a single Pareto optimal region, optimization based on non-conditional sampling is often sufficient, since there is less need to assume that the currently identified Pareto front is not globally optimal.

We evaluate the performance of different acquisition strategies on multiple benchmark problems characterized by complex Pareto optimal solution distributions. The first is a simulated bi-objective Gaussian mixture (GM; Fröhlich et al. (2020)) function with a 2-dimensional design space, where the Pareto optimal solutions are widely dispersed. The second is the car side-impact problem (RE4-7-1), which involves a 7-dimensional design space and four objectives, with an unknown Pareto optimal set (Tanabe & Ishibuchi, 2020). The third one is Penicillin composition inverse design problem (Liang & Lai, 2021). All experiments are implemented within the BOTorch framework (Balandat et al., 2020). Additional implementation details are provided in Appendix A.3.

For each problem, we evaluate optimization performance using two metrics. The first is hypervolume, which measures the quality of the solutions obtained in the objective space. The second is expected minimum distance (EMD), defined in (12), which quantifies the coverage of the Pareto optimal solutions in the design space. EMD is conceptually similar to the IGD, but it operates in the design space rather than the objective space. In MOBO, fully covering the Pareto front does not guarantee that all Pareto optimal designs are recovered. However, capturing all Pareto optimal designs does imply full coverage of the Pareto front. Therefore, EMD serves as a stricter metric than IGD, offering higher selectivity and precision in evaluating MOBO solution quality:

$$\text{EMD}(\mathcal{X}_n, \mathcal{X}^*) = \frac{1}{|\mathcal{X}^*|} \sum_{\mathbf{x}^* \in \mathcal{X}^*} \min_{\mathbf{x} \in \mathcal{X}_n} \|\mathbf{x} - \mathbf{x}^*\|_2. \quad (12)$$

As shown in Figures 3, 4, and 5, POMH consistently outperforms all competing baselines in terms of hypervolume across all three problems. In addition, POMH achieves strong diversity performance, as measured by EMD on the

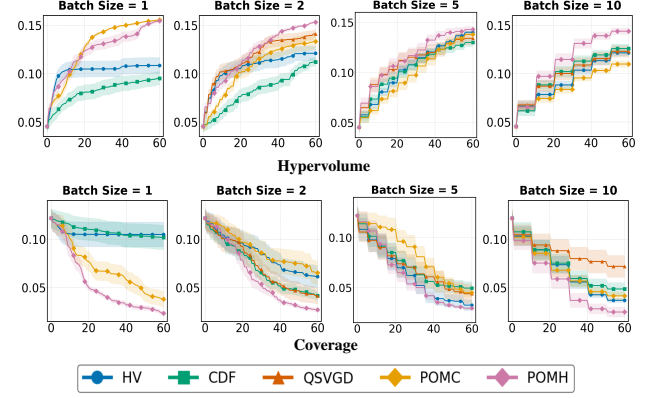


Figure 3. Comparison of batch MOBO performances on the GM problem across different acquisition strategies: qEHVI, QSVGD, CDF, POMC and POMH.

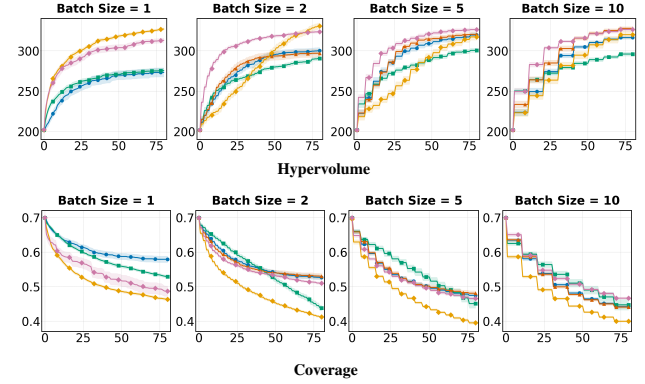


Figure 4. Comparison of batch MOBO performances on RE4-7-1 problem across different acquisition strategies: qEHVI, QSVGD, CDF, POMC and POMH. Legend is the same as Figure 3.

GMM problem. The effect of batch size is relatively minor for POMH across these settings. POMC demonstrates favorable diversity performance on the RE4-7-1 and Penicillin problems, while its hypervolume performance remains competitive, particularly for smaller batch sizes. In contrast, qEHVI and CDF exhibit degraded performance when the batch size is small, primarily due to their tendency to over-sample local optima or extreme value regions. As the batch size increases, this issue is partially mitigated, leading to improved performance. Notably, this batch size sensitivity is not observed for the POM-based methods, which indicates they can avoid excessive sampling in local optimal regions even under small batch sizes. While QSVGD can achieve reasonable performance with careful hyperparameter tuning, however it is outperformed by POMH across most scenarios.

In addition to the detailed performance analysis on the three synthetic problems, we also conducted experiments on several other benchmark tasks, including RE3-5-4 (Tanabe & Ishibuchi, 2020) and the widely used DTLZ and ZDT test subjects. The corresponding results are provided in Appendix A.4. Despite the popularity of DTLZ and ZDT

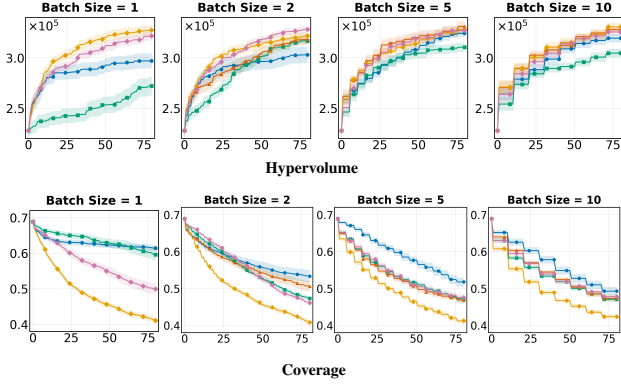


Figure 5. Comparison of batch MOBO performances on Penicillin problem across different acquisition strategies: qEHVI, QSVGD, CDF, POMC and POMH. Legend is the same as Figure 3.

benchmarks, we observe that their Pareto optimal solutions are typically either continuously connected within a single region or heavily concentrated near boundary areas, as illustrated in Figures 7 and 8. Both cases can be regarded as effectively unimodal objective landscapes that require relatively little diversity capability. Therefore, we do not consider these benchmarks particularly suitable for evaluating diversity performance.

Meanwhile, we also provide additional results on a high-dimensional, many-objective task, namely the General Aviation Aircraft (GAA) problem (Simpson et al., 1996), which involves 27 input variables and 9 objectives. Similar to several MOBO baselines, including QSVGD and qEHVI, this problem suffers significantly from the curse of dimensionality due to its large search space. In addition, POMH employs a hypervolume computation strategy similar to that used by qEHVI in BOTorch. Consequently, completing experiments for qEHVI, QSVGD, and POMH on the GAA problem is computationally challenging. Therefore, Appendix A.4 reports only partial results, including POMC and other methods that are less affected by the high-dimensional setting.

As POM is decomposed into several estimation components, we conduct ablation studies to investigate the contribution of each component individually. Specifically, we compare the performance of different metrics and their combinations, including Pareto optimality (PO), coverage based methods (qEHVI and CDF), Pareto optimal region sampling (PORS), POMC, and POMH, on the three synthetic benchmark functions. Detailed ablation results are provided in Appendix A.5.

4.3. Real-world case studies

We further evaluate and compare batch MOBO performance in an alloy inverse design task aimed at identifying compositions with targeted material properties. Specifically, we aim to recover Pareto optimal compositions from a pool

of 1,000 given candidates under a constrained evaluation budget (Hastings et al., 2024). To provide a comprehensive assessment on the effectiveness of different batch MOBO strategies, we vary the batch size and measure the resulting Pareto optimal solution rediscovery ratio, for qEHVI, QSVGD, and POM.

The design objectives in this MOBO problem include six material properties: Stacking Fault Energy (SFE), principal axial elastic constant (C_{11}), Heat Capacity (HC), Thermal Conductivity (TC), Solidification Range (SR), and Room-Temperature Density (RTD). As a model of the trade-offs inherent to alloy design, these six material properties: SFE, C_{11} , HC, TC, SR, and RTD, collectively define the key thermodynamic, mechanical, and processing constraints that shape performance. These attributes are deeply entangled: they do not vary independently, and optimizing one often perturbs others. Their interplay reflects the practical complexity of real-world alloy development, where performance gains are rarely free and improvements in one dimension often impose costs elsewhere.

SFE governs dislocation behavior, including splitting, cross-slip, and twinning, and directly influences deformation mechanisms such as transformation-induced plasticity (TRIP) and twinning-induced plasticity (TWIP) (Gallagher, 1970; Ahlers, 1970; Vitos et al., 2006; Pierce et al., 2015). C_{11} characterizes the intrinsic stiffness along the primary crystallographic axes and plays a key role in determining both the yield strength and elastic anisotropy (Hill, 1948; Yoo, 1986; Hutchinson, 2015). Thermal properties fall into two regimes: HC, which governs transient thermal response by storing or releasing energy; and TC, which dictates steady-state heat transport by controlling spatial heat diffusion (Callister & Rethwisch, 2013). Both properties are critical in thermally demanding systems, from aerospace components to microelectronic devices (Hanuska et al., 2000; Zawada, 2006). These thermal properties often exhibit an inverse relationship with RTD, a measure of mass per unit volume. While lower density can enhance specific properties, it may compromise absolute stiffness, thermal conductivity, or mechanical robustness (Ashby & Cebon, 2005; Callister & Rethwisch, 2013). SR reflects the temperature interval over which an alloy transitions from liquid to solid. A narrow SR minimizes elemental segregation and reduces the risk of hot cracking during solidification, thereby enhancing processability; a wider SR expands the accessible compositional design space but increases susceptibility to solidification defects (Flemings, 1974).

Simultaneously optimizing all six dimensions is inherently constrained by competing material mechanisms. For example, alloying strategies to increase ductility via reduced SFE or enhance stiffness via increased C_{11} may inadvertently raise density or broaden SR, thus worsening processability

or reducing mass efficiency (Ashby, 2011c;a;b). These trade-offs cannot be resolved independently; rather, they manifest as interlocked constraints in multi-objective optimization landscapes. The alloy design process must therefore navigate a multi-dimensional Pareto front, where gains in one domain often exact cost in another, a reality mirrored in recent MOBO frameworks that explicitly balance candidate quality with design space coverage (Khatamsaz et al., 2021; 2022).

Here, we study six constructed MOBO tasks by grouping six properties into three bi-objective (Bi-1: C_{11} and SFE, Bi-2: HC and TC, Bi-3: SR and RTD), two tri-objective (Tri-1: RTD, C_{11} , and SFE, Tri-2: SR, HC, and TC), and all six objective tasks (All: SR, RTD, HC, TC, C_{11} , and SFE). We use the following metrics to evaluate the re-identification performance: 1) Rediscovery ratio; 2) Hypervolume; 3) EMD; 4) IGD; 5) Maximum Spread (Zitzler & Thiele, 2002); and 6) Spacing (Schott, 1995). Rediscovery ratio is defined as the number of rediscovered Pareto optimal solutions divided by the total number of true Pareto optimal solutions. It serves as the most practical evaluation metric for materials inverse design, as it directly measures the effectiveness of recovering target compositions. Therefore, we adopt it as the primary metric for comparison. Figure 6 presents the rediscovery ratios achieved by different acquisition strategies across various experimental setups. The evaluation results for other metrics and the acquisition behavior analysis exhibit consistent trends with those observed in the benchmark examples (Appendix A.6).

For all the tasks, performing 80 random queries from 1000 candidates has a probability of 0.08 for any Pareto optimal solution being selected. Each method outperforms this baseline of 0.08. As the number of objectives increases, the rediscovery ratio decreases as expected, which is illustrated when comparing the trends in Figures 6a, 6d, and 6f; or similarly in Figures 6b, 6e, and 6f when increasing the number of objectives under consideration from two, three, to six.

Specifically, POM based acquisitions consistently rediscovers the most Pareto optimal solutions compared to qEHVI and QSVGD across different settings. It typically shows stable performance with a smaller batch size, although on task Tri-2 this is not always the case, as Figure 6e shows. For qEHVI, small batch sizes lead to low acquisition efficiency on tasks such as bi-objective (C_{11} and SFE; HC and TC; SR and RTD) and tri-objective (RTD, C_{11} , and SFE) tasks. Performance improves with larger batch sizes, as the coverage probability $P(\mathcal{X}^* \subseteq \mathbf{X} \mid \mathbf{X} \subseteq \mathcal{X}^*)$ naturally increases. However, larger batches can reduce the probability $P(\mathbf{X} \subseteq \mathcal{X}^*)$, causing an overall performance drop as seen in Figure 6f.

For QSVGD, we employ a decaying schedule for the hyperparameter η to gradually diminish the influence of the

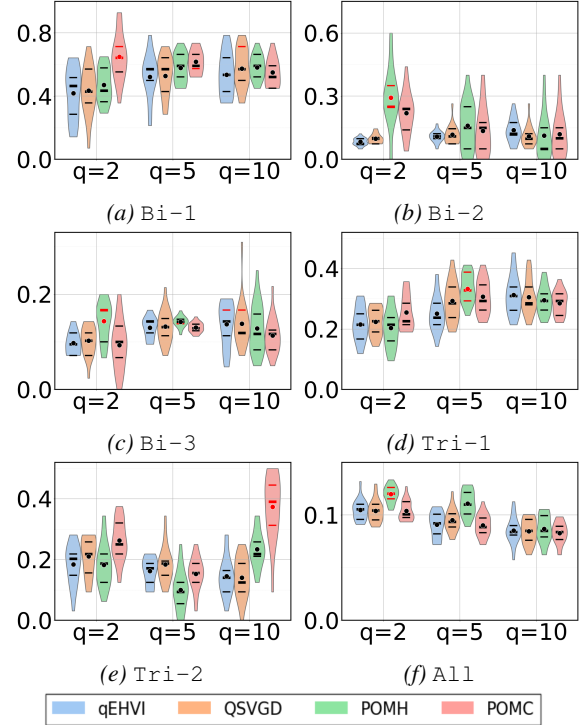


Figure 6. Rediscovery performance comparison for different MOBO settings: bi-objective combinations considering (a) Bi-1: C_{11} and SFE, (b) Bi-2: HC and TC, (c) Bi-3: SR and RTD; tri-objective tasks considering (d) Tri-1: RTD, C_{11} , and SFE, (e) Tri-2: SR, HC, and TC; as well as considering all six objectives: (f) All: SR, RTD, HC, TC, C_{11} , and SFE. Dots: Mean value across 20 trials; Red: Best performance across all different settings.

entropy term over iterations, thereby achieving a balance between exploitation and diversity (details in Appendix A.3). Finding the optimal exploration-exploitation balance remains challenging, as it varies by different scenarios. Without a dynamic balance, the diversity term may occasionally dominate the qEHVI term, causing the acquisition of candidates without any Pareto front potential. Consequently, QSVGD’s overall performance is worse than POM and sometimes can even be worse than qEHVI.

5. Conclusion and Limitation

We propose a novel batch acquisition strategy for MOBO that directly maximizes the probability that the acquired batch candidates match the true Pareto optimal set. In contrast to prior approaches, where acquisition indicators are only indirectly related to the optimization objective, POM formulation provides a theoretical definition of this probability and a principled strategy for estimating it by restricting sampling to regions that are potentially Pareto optimal. We evaluate POM on a range of challenging real-world benchmark problems and further demonstrate its practical effectiveness in an alloy discovery application.

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Impact Statement

This paper presents work studying batch multi-objective Bayesian optimization. There are many potential societal consequences of our work when multiple black-box optimization criteria need to be considered, which is typical real-world settings involving different experimental design and/or decision making. The presented research focuses on algorithm development and we feel that it has diverse applications but none of them must be specifically highlighted here.

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A. Appendix

A.1. Indicator selection and discussion

Hypervolume is one of the most widely used coverage indicators. Comparing *coverage* with the hypervolume definition reveals strong conceptual similarities. The primary distinctions are that hypervolume is defined directly in the objective space and depends on a user specified reference point. By assigning uniform weight across the objective space, hypervolume naturally prefers regions of the Pareto front with sparse coverage, thereby encouraging exploitation in areas that are difficult to acquire through screening the design space. Despite its advantages, hypervolume also exhibits notable limitations. Since the true distribution of objective values in the objective space is generally unknown, selecting an appropriate reference point is necessary but nontrivial. At present, there is limited theoretical guidance for reference point selection, making this choice largely heuristic and often dependent on prior knowledge, which may introduce bias toward particular regions of the objective space. Moreover, the assumption of a uniform distribution over objective values can lead to inaccurate dominance estimates, as portions of the rectangular region defined by the reference point may not contain any feasible or attainable objective values. Consequently, hypervolume based dominance measures may account for objective vectors that do not correspond to any realizable solutions.

Another indicator, CDF, estimates coverage by directly sampling candidate solutions from the design space. For each candidate, we compare its expected objective values against those of other sampled candidates and count how many of them are dominated. This approach ensures that each dominance count corresponds to realizable objective vectors, thereby avoiding the inclusion of infeasible or nonexistent regions of the objective space. A limitation of this approach is that it is infeasible to exhaustively sample the design space, particularly in regions where the objective landscape changes sharply. Failure to sample such regions may incorrectly suggest that certain objective values are unattainable. As a result, gaps in sparsely populated regions of the Pareto front are difficult to cover, and the acquired Pareto front tends to reflect the true underlying Pareto front density rather than a uniformly distributed one. In our setup, we have implemented both indicators to estimate the *Coverage* performance.

A.2. Computational Efficiency

We validate the computational complexity derived in Section 3.3 by investigating the run-time of different batch MOBO strategies in our alloy design case study. Table 1 reports the average runtime required to generate one evaluation candidate across different acquisition strategies and experimental settings. The runtime is mainly influenced by the batch size q and the number of objectives m . In general, incorporating Pareto optimality estimation introduces minimal additional overhead compared to coverage estimation. For instance, with multiple objectives ($m = 6$), the cost is dominated by hypervolume estimation (Yang et al., 2019), particularly for large q , since $(n + q)^2 m \ll (2^q - 1)m$. For POM, the computational cost exhibits a higher standard deviation, likely due to unstable Pareto front numbers as the n increases. In contrast, the other two strategies demonstrate more stable performance across trials. Meanwhile, the actual computational cost may deviate from the theoretical complexity analysis, as some acquisition optimization converges to the maximum before exhausting the optimization budget.

Table 1. Runtime (in seconds, Average $\pm$ Standard Deviation) per candidate evaluation for different tasks, batch sizes, and BO strategies

Acquisition strategies	Batch size	Bi-1	Bi-2	Bi-3	Tri-1	Tri-2	All
qEHVI	2	3.37 $\pm$ 1.17	4.50 $\pm$ 1.70	1.64 $\pm$ 0.56	5.15 $\pm$ 0.87	6.22 $\pm$ 4.12	9.30 $\pm$ 5.05
	5	4.72 $\pm$ 2.35	2.44 $\pm$ 1.30	2.28 $\pm$ 0.75	5.79 $\pm$ 2.01	6.77 $\pm$ 0.63	46.03 $\pm$ 52.18
	10	7.01 $\pm$ 2.82	4.25 $\pm$ 0.30	4.24 $\pm$ 1.42	11.50 $\pm$ 6.93	12.97 $\pm$ 0.37	60.09 $\pm$ 26.58
QSVGD	2	6.69 $\pm$ 0.86	4.07 $\pm$ 0.59	3.28 $\pm$ 1.81	7.74 $\pm$ 1.77	8.97 $\pm$ 3.06	15.22 $\pm$ 5.50
	5	3.88 $\pm$ 1.30	7.32 $\pm$ 1.62	2.35 $\pm$ 1.55	9.97 $\pm$ 1.71	11.80 $\pm$ 0.54	56.23 $\pm$ 57.17
	10	9.45 $\pm$ 0.92	5.09 $\pm$ 0.73	5.09 $\pm$ 1.85	13.72 $\pm$ 6.83	15.20 $\pm$ 0.49	72.79 $\pm$ 49.03
POMH	2	6.85 $\pm$ 0.90	4.15 $\pm$ 0.65	3.45 $\pm$ 1.90	7.95 $\pm$ 1.85	9.15 $\pm$ 3.10	15.50 $\pm$ 5.80
	5	4.05 $\pm$ 1.40	7.50 $\pm$ 1.70	2.55 $\pm$ 1.60	10.15 $\pm$ 1.80	12.10 $\pm$ 0.60	57.10 $\pm$ 58.20
	10	9.65 $\pm$ 1.05	5.25 $\pm$ 0.80	5.25 $\pm$ 1.95	13.95 $\pm$ 7.05	15.55 $\pm$ 0.55	73.50 $\pm$ 50.15
POMC	2	6.95 $\pm$ 0.95	4.25 $\pm$ 0.70	3.60 $\pm$ 2.00	8.10 $\pm$ 1.95	9.35 $\pm$ 3.25	15.85 $\pm$ 6.10
	5	4.20 $\pm$ 1.55	7.65 $\pm$ 1.85	2.70 $\pm$ 1.75	10.40 $\pm$ 2.00	12.45 $\pm$ 0.75	58.25 $\pm$ 59.40
	10	9.80 $\pm$ 1.15	5.40 $\pm$ 0.95	5.40 $\pm$ 2.10	14.25 $\pm$ 7.30	15.90 $\pm$ 0.70	74.85 $\pm$ 51.60

A.3. Experimental Setup and Environment

We consider the task of identifying Pareto optimal solutions from a discrete feasible set of 10,000 candidates for all three benchmark problems mentioned in the main text. All the evaluation experiments are implemented in the `BOToRch` framework (Balandat et al., 2020). Given a strict evaluation budget of 80 (60 for GM problem) function queries per trial, each BO run is initialized with 10 randomly selected points. The corresponding GP surrogate models are trained using these initial evaluations. For each batch selection, we evaluate the acquisition function using MC sampling with 512 samples to ensure accurate estimation. The adopted optimization strategy for each iteration in `BOToRch` is set to be *joint* instead of *sequential* to further increase the estimation accuracy. Further details on the distinction between the *joint* and *sequential* batch selection strategies can be found in the prior work (Ament et al., 2023), which is adopted in the `BOToRch` framework.

For QSVGD, we assign equal weights to all quantiles, as none of our experimental setups incorporate a notion of risk. For the hyperparameter η , we define a piecewise schedule for the hyperparameter η : it is kept constant at η_0 for the first 40 (30 for GM problem) iterations to maintain exploration, and then linearly decays to 0 over the next 40 (30 for GM problem) iterations, effectively reducing the influence of the entropy term as optimization progresses. The choice of η_0 is task-dependent and is generally tuned so that the weighted entropy term is on a similar scale as the hypervolume improvement term, ensuring neither heavily dominates the optimization objective.

We compute the hypervolume using the default reference point, defined as 1.1 times the lower bounds of the objectives, for all three benchmark problems. The exact hypervolume is computed using the box decomposition method (Daulton et al., 2020). To compute EMD, we first exhaustively evaluate the objective values of all 10,000 candidates in the feasible space to obtain the true Pareto optimal solutions. We then compute EMD between the current non-dominated solutions and the true Pareto optimal solutions as a measure of convergence quality.

For all objective functions, the experimental runs were distributed across 8 Intel(R) Xeon(R) Gold 6248R CPUs.

A.4. Additional synthetic benchmark results

EMMI is an objective-space filling method that aims to improve the diversity and uniformity of the Pareto front (PF). EHVI-SF (EHVI with space filling) incorporates space-filling strategies in the design space to enhance diversity. Parallel Pareto Efficient Global Optimization (qParEGO Knowles (2006)) is one of the most classic approaches, which performs MOBO by scalarizing multiple objectives into a single objective. Multi-objective Joint entropy search Tu et al. (2022) focuses on reducing uncertainty over the Pareto optimal solutions. qPOTS achieves Pareto optimality via Thompson Sampling while promoting diversity through space filling in the design space. DGEMO improves diversity by partitioning the space and selecting representative solutions. Hypervolume knowledge gradient (HVKG Daulton et al. (2023)) is a knowledge-gradient method based on hypervolume; it is a one-step-look-ahead policy and is typically used in settings with decoupled objective evaluations where objectives cannot be evaluated simultaneously.

We also added three additional synthetic problems: RE354, DTLZ3, and ZDT3. RE354 is a 5-dimensional problem with 3 objectives, while the DTLZ and ZDT families are classic MOBO benchmark test functions. The experimental setup for these 3 additional problems are the same as the GMM problem mentioned in the main text. The additional batch performance analysis is mainly conducted on RE354. We present all results together to provide a clearer overall comparison. QSVGD is not included, since there is no entropy term when the batch size is 1.

Table 2. Hypervolume for different synthetic functions

Problem	qParEGO	qJES	qEMMI	qEHVI	qEHVI-SF	qCDF	POMH	POMC	qPOTS	DGEMO	qHVKG
GMM	0.108	0.145	0.132	0.109	0.149	0.932	0.156	0.156	0.133	0.122	0.115
RE471	278	330	281	271	329	269	325	336	298	289	293
Penicillin	2.65	2.74	3.16	2.95	3.20	2.72	3.42	3.33	3.20	2.82	2.96
RE354	162	197	177	169	203	165	211	198	202	177	182
DTLZ3	1.02	1.17	1.50	1.40	1.46	1.37	1.49	1.42	1.48	1.24	1.35
ZDT3	15.2	17.3	20.1	19.9	21.0	19.6	21.0	19.5	21.6	18.2	20.0

For complex PFs like the three main-text problems and RE354, POMH shows strong hypervolume, and POMC performs well on EMD. For simpler cases like DTLZ (connected PF) and ZDT (optima near boundaries), POMH/POMC still perform well but do not dominate, as these functions need little diversity, making POM’s diversity term unnecessary. Batch results follow the same trend; POMH/POMC are unaffected by batch size.

Table 3. EMD for different synthetic functions

Problem	qParEGO	qJES	qEMMI	qEHVI	qEHVI-SF	qCDF	POMH	POMC	qPOTS	DGEMO	qHVKG
GMM	0.092	0.031	0.087	0.105	0.029	0.103	0.013	0.037	0.050	0.033	0.082
RE471	0.617	0.502	0.576	0.585	0.524	0.536	0.489	0.457	0.500	0.515	0.512
Penicillin	0.654	0.583	0.428	0.612	0.437	0.591	0.498	0.413	0.444	0.487	0.585
RE354	1.13	0.432	0.641	0.709	0.310	0.725	0.326	0.348	0.495	0.394	0.603
DTLZ3	0.872	0.571	0.776	0.790	0.538	0.809	0.501	0.484	0.504	0.570	0.742
ZDT3	0.981	0.617	0.782	0.716	0.590	0.873	0.601	0.588	0.612	0.662	0.703

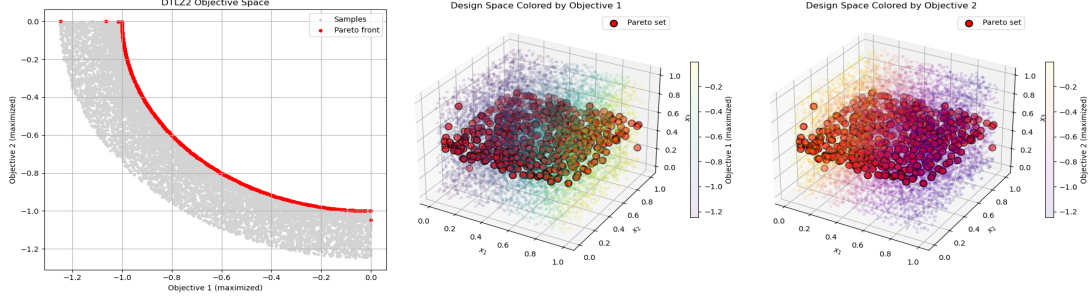


Figure 7. Pareto optimal solution distribution of DTLZ2: The left plot shows the Pareto front in the objective space. The middle and right plots illustrate the distribution of Pareto optimal solutions in the design space, where each plot corresponds to one objective. The color bar indicates the respective objective function values.

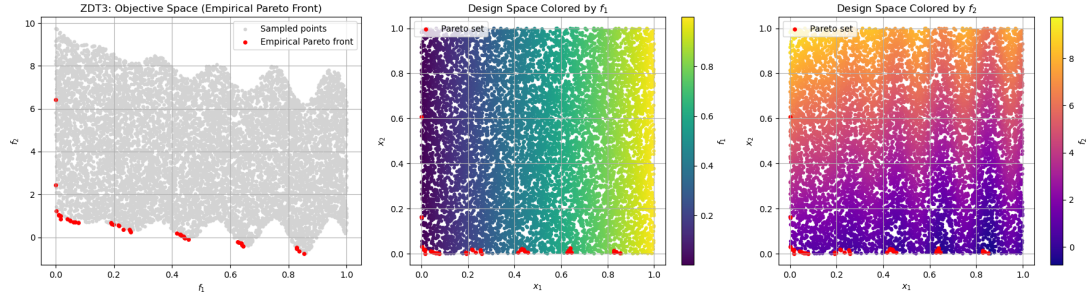


Figure 8. Pareto optimal solution distribution of ZDT2: The left plot shows the Pareto front in the objective space. The middle and right plots illustrate the distribution of Pareto optimal solutions in the design space, where each plot corresponds to one objective. The color bar indicates the respective objective function values.

Table 4. Batch size effect on HV for problem RE354

q	qParEGO	qJES	qEMMI	qEHVI	qEHVI-SF	qCDF	POMH	POMC	qPOTS	DGEMO	qHVKG
1	162	197	177	169	203	165	211	198	203	175	182
2	166	193	182	178	205	179	212	210	201	194	187
5	170	189	172	188	202	201	200	190	193	193	182

Table 5. Batch size effect on EMD for problem RE354

q	qParEGO	qJES	qEMMI	qEHVI	qEHVI-SF	qCDF	POMH	POMC	qPOTS	DGEMO	qHVKG
1	1.13	0.432	0.641	0.709	0.310	0.725	0.326	0.348	0.475	0.401	0.600
2	0.93	0.437	0.592	0.683	0.308	0.732	0.314	0.302	0.465	0.374	0.584
5	1.02	0.500	0.517	0.667	0.312	0.674	0.337	0.311	0.450	0.354	0.436

In high-dimensional problems, the coverage indicator tends to dominate the behavior of the acquisition process, while Pareto optimality becomes less influential because objective vectors are often nearly orthogonal, making dominance relationships

difficult to establish. As shown in Tables 6 and 7, the POMC method consistently outperforms other methods across different noise levels.

Table 6. HV performance (10^{23}) for problem GAA

SNR	CDF	POMC	RS	qPOTS
1	1.325	1.349	1.311	1.326
2	1.564	1.672	1.365	1.425
4	1.637	2.010	1.351	1.359
8	1.718	2.346	1.293	1.307

Table 7. EMD performance for GAA

SNR	CDF	POMC	RS	qPOTS
1	0.868	0.845	0.889	0.910
2	0.805	0.726	0.873	0.825
4	0.797	0.570	0.852	0.832
8	0.730	0.514	0.891	0.799

A.5. Ablation study

The ablation studies highlight the contribution of each component. PO is primarily used to construct a backbone of the Pareto front; when incorporated into estimators such as CDF or hypervolume based methods, it accelerates the discovery of extreme Pareto optimal solutions. Since CDF tends to focus more on high density, easily sampled regions, PO provides a more pronounced improvement for CDF compared to EHVI. PORS is designed to mitigate local trapping by assuming that the current Pareto optimal set is the true Pareto front. It improves performance for both EHVI and CDF, helping them escape local optima and yielding larger performance gains than PO alone.

Table 8. POMC ablation HV

Problem	CDF	CDF+PO	CDF+PORS	POMC
GMM	0.932	0.105	0.133	0.156
RE471	269	301	289	336
Penicillin(10^5)	2.72	3.05	3.27	3.42

Table 9. POMH ablation HV

Problem	HV	HV+PO	HV+PORS	POMH
GMM	0.109	0.113	0.142	0.156
RE471	271	278	319	325
Penicillin(10^5)	2.95	3.06	3.29	3.33

Table 10. POMC ablation EMD

Problem	CDF	CDF+PO	CDF+PORS	POMC
GMM	0.103	0.086	0.0532	0.037
RE471	0.536	0.501	0.472	0.457
Penicillin(10^5)	0.591	0.555	0.542	0.413

Table 11. POMH ablation EMD

Problem	HV	HV+PO	HV+PORS	POMH
GMM	0.105	0.073	0.020	0.013
RE471	0.585	0.546	0.497	0.489
Penicillin(10^5)	0.612	0.604	0.521	0.498

A.6. Additional Evaluation Metrics for Alloy Inverse Design

In our materials inverse design tasks, for each property, we first train a property predictor on the full candidate set and use this surrogate model as the black-box objective function. We initialize each batch MOBO run with 10 randomly selected compositions and allocate a total of 80 evaluations. The goal is to compare acquisition strategies in terms of how many Pareto optimal solutions they are able to recover within this evaluation budget. We vary the batch size from the set 2, 5, 10, corresponding to 40, 16, 8 batch MOBO iterations. In addition to the rediscovery ratio discussed in the main text, we also evaluate the performance of different acquisition strategies using several commonly adopted multi-objective evaluation metrics. These include IGD (Ishibuchi et al., 2015), Maximum Spread (Zitzler & Thiele, 2002), and Spacing (Schott, 1995).

Here, we introduce three additional performance metrics in addition to the ones discussed in the main text: IGD, Maximum Spread (MS), and Spacing (SP). IGD assesses the coverage of the Pareto front in the objective space. However, since it operates in the target objective space and multiple Pareto optimal solutions can be mapped to the same objective values, IGD may be considered a weaker indicator of diversity compared to EMD. MS measures whether the extreme value regions of the Pareto front are captured by the selected candidates. SP evaluates the uniformity of the batch distribution, reflecting how evenly the solutions are spread across the Pareto front. The definitions of these three additional evaluation metrics can be found in (13), (14), and (15):

$$\text{IGD}(\mathcal{Y}_n^*, \mathcal{Y}^*) = \frac{1}{|\mathcal{Y}^*|} \sum_{\mathbf{y} \in \mathcal{Y}^*} \min_{\mathbf{y}' \in \mathcal{Y}_n^*} \|\mathbf{y} - \mathbf{y}'\|_2, \quad (13)$$

$$\text{MS}(\mathcal{Y}_n^*) = \sqrt{\frac{1}{m} \sum_{k=1}^m \left(\max_{i=1, \dots, |\mathcal{Y}_n^*|} y_k^{(i)} - \min_{i=1, \dots, |\mathcal{Y}_n^*|} y_k^{(i)} \right)^2}, \quad (14)$$

$$\text{SP}(\mathcal{Y}_n^*) = \sqrt{\frac{1}{|\mathcal{Y}_n^*| - 1} \sum_{i=1}^{|\mathcal{Y}_n^*|} (d_i - \bar{d})^2}, \quad (15)$$

where

$$\mathcal{Y}_n^* = \{\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(|\mathcal{Y}_n^*|)}\}, \quad \mathbf{y}^{(i)} = (y_1^{(i)}, y_2^{(i)}, \dots, y_m^{(i)}) \in \mathbb{R}^m$$

$$d_i = \min_{j=1, \dots, |\mathcal{Y}_n^*|, j \neq i} \|\mathbf{y}^{(i)} - \mathbf{y}^{(j)}\|_1, \quad \bar{d} = \frac{1}{|\mathcal{Y}_n^*|} \sum_{i=1}^{|\mathcal{Y}_n^*|} d_i.$$

For performance comparison by IGD, shown in Table 12, MS, shown in Table 13 and SP, shown in Table 14. POM demonstrates superior performance compared to other acquisition strategies. This indicates that, relative to qEHVI and QSVGD, POM not only achieves better coverage of the objective space and rediscovery of more extreme solutions but also produces a more uniformly distributed Pareto front. Performance comparison results with these three metrics provide strong complementary evidence that POM recovers the Pareto front more effectively and reinforce the conclusion drawn from the reported results on the rediscovery rate in the main text.

Table 12. Comparison of Pareto front coverage with respect to IGD↓

Acquisition strategies	q	Bi-1	Bi-2	Bi-3	Tri-1	Tri-2	All
qEHVI	2	38.74±30.78	4.21±2.37	18.60±11.63	30.77±22.01	19.71±7.00	45.11±2.43
	5	28.84±14.99	3.51±2.05	14.55±8.50	23.65±10.68	20.21±4.87	45.46±2.41
	10	26.23±10.77	2.92±2.17	10.07±5.88	17.57±6.51	25.23±8.48	43.62±1.45
QSVGD	2	37.41±31.06	3.86±2.56	15.83±9.34	27.49±18.60	18.94±3.70	45.57±2.56
	5	25.40±13.71	3.49±1.75	12.23±5.88	21.17±8.53	23.23±7.34	44.43±2.40
	10	19.54±10.14	4.05±2.03	14.00±9.78	18.74±7.10	26.41±10.97	43.30±1.68
POMH	2	17.22±12.67	2.88±1.00	8.89±5.50	15.68±8.31	18.19±5.59	41.92±1.83
	5	14.75±6.70	2.11±0.68	8.86±4.98	13.14±7.42	16.11±6.40	42.21±2.44
	10	19.14±8.60	3.52±2.96	10.36±6.17	16.26±7.28	21.01±9.35	40.64±1.33
POMC	2	27.31±12.45	3.37±1.08	12.36±5.63	21.59±8.44	18.57±5.41	43.75±1.91
	5	20.07±6.84	2.80±0.72	10.54±5.11	17.16±7.35	19.67±6.58	43.32±2.37
	10	19.34±8.48	3.79±3.14	12.18±6.05	17.50±7.56	23.71±9.62	41.97±1.41

Table 13. Comparison of extreme Pareto capturing with respect to Maximum Spread (MS)↑

Acquisition strategies	q	Bi-1	Bi-2	Bi-3	Tri-1	Tri-2	All
qEHVI	2	876.2±112.7	67.6±8.3	431.5±170.1	956.6±166.7	525.1±115.0	968.9±88.4
	5	944.9±186.8	70.6±8.9	465.1±155.2	1010.4±173.6	512.1±100.2	972.0±138.7
	10	953.6±135.0	74.3±9.1	515.1±127.4	1055.1±162.8	460.6±129.2	1006.2±144.3
QSVGD	2	911.5±108.1	71.1±9.0	427.5±153.3	1075.8±237.2	483.4±92.8	984.8±147.9
	5	994.3±167.3	71.4±7.2	467.8±126.2	1047.6±145.8	455.6±120.9	1008.1±137.1
	10	968.0±144.8	67.8±9.6	454.8±154.2	1087.4±145.5	459.5±146.3	1033.0±147.3
POMH	2	1031.5±161.8	74.2±5.9	528.1±129.8	1102.2±122.3	558.1±102.3	992.1±122.6
	5	1152.9±179.9	77.4±3.7	532.1±112.4	1165.4±270.8	562.2±111.4	1015.7±152.2
	10	988.2±109.3	72.0±10.2	492.4±128.4	1022.2±146.7	488.9±145.0	1041.2±147.5
POMC	2	1028.9±160.9	74.0±5.7	526.3±130.6	1105.4±123.1	556.7±101.5	989.5±121.8
	5	1148.2±181.1	78.1±3.8	521.8±111.6	1172.1±269.9	545.9±112.2	1021.3±151.4
	10	992.5±108.7	72.1±10.0	498.2±129.1	1035.6±145.9	492.3±144.2	1038.5±148.1

Table 14. Comparison of Pareto front uniformity with respect to Spacing (SP)↓

Acquisition strategies	q	Bi-1	Bi-2	Bi-3	Tri-1	Tri-2	All
qEHVI	2	1.19±0.09	1.35±0.13	1.42±0.10	1.38±0.09	1.12±0.21	0.75±0.04
	5	1.08±0.10	1.13±0.11	1.33±0.09	1.25±0.10	0.99±0.18	0.77±0.07
	10	0.99±0.08	1.04±0.10	1.28±0.11	1.11±0.09	0.74±0.14	0.67±0.04
QSVGD	2	1.12±0.11	1.27±0.10	1.36±0.09	1.28±0.10	1.05±0.13	0.75±0.03
	5	1.14±0.07	1.18±0.10	1.33±0.08	1.24±0.09	0.85±0.17	0.73±0.05
	10	1.03±0.09	1.05±0.08	1.24±0.12	1.09±0.07	0.73±0.09	0.68±0.04
POMH	2	0.74±0.06	1.02±0.18	1.22±0.12	1.10±0.11	1.15±0.14	0.67±0.05
	5	0.68±0.06	0.84±0.11	1.13±0.13	0.91±0.08	0.82±0.07	0.66±0.04
	10	0.62±0.05	0.70±0.09	1.07±0.11	0.75±0.10	0.65±0.09	0.63±0.06
POMC	2	0.75±0.07	1.04±0.17	1.24±0.11	1.12±0.10	1.16±0.13	0.68±0.05
	5	0.69±0.05	0.86±0.10	1.15±0.12	0.93±0.07	0.84±0.06	0.67±0.05
	10	0.64±0.06	0.72±0.10	1.09±0.13	0.77±0.09	0.64±0.10	0.62±0.05

A.7. Use of Large Language Models

Large Language Models are only used to check vocabulary and grammar for polishing purpose.